

Lesson 05 · From average slope to the derivative

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Find the slope of the straight line passing through $(2, 5)$ and $(4, 9)$.

☐ A $\frac{1}{2}$

☐ B 2

☐ C 4

☐ D -2

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Q2 · Rank 1 / Foundation

Find the slope of the straight line passing through $(3, -1)$ and $(7, -5)$.

☐ A 1

☐ B -4

☐ C -1

☐ D 4

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Q3 · Rank 1 / Foundation

Find the slope of the straight line passing through $(7, 19)$ and $(4, 19)$.

☐ A Undefined

☐ B -3

☐ C 19

☐ D 0

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Q4 · Rank 2 / Core

Differentiate the function $f(x) = x^7$.

☐ A $7x^6$

☐ B x^6

☐ C $7x^7$

☐ D $x^8/8$

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Q5 · Rank 2 / Core

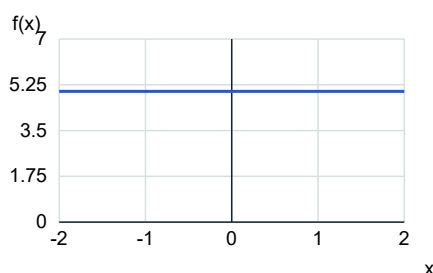
Differentiate $y = x^8$.

☐ A $8x^8$
☐ B $8x^7$
☐ C x^7
☐ D $x^9/9$

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Q6 · Rank 2 / Core

Sketch the graph of the function $f(x) = 5$. Explain why it follows from this that $f'(x) = 0$. Choose the explanation.


☐ A The graph is vertical.

☐ B Its height is zero.

☐ C The graph is horizontal, so every tangent has zero slope.

☐ D Its slope is 5.

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Q7 · Rank 3 / Application

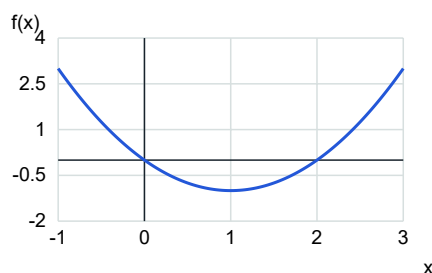
Differentiate the function $f(x) = x^7$. Hence calculate the slope of the graph of $y = x^7$ at the point $x = 2$.

☐ A 128☐ B 896☐ C 14☐ D 448

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Q8 · Rank 3 / Application

For the function $f(x) = x^2 - 2x$, sketch the graph and, by measuring the slope of the tangents, estimate $f'(1)$.

☐ A 0☐ B -1☐ C 2☐ D 1

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Q9 · Rank 3 / Application

For the graph $y = \sqrt{x}$, A is the point where $x = 4$, and B is the point where $x = 4.1$. Find the gradient of the chord AB.

☐ A $\frac{1}{4}$

☐ B $\frac{\sqrt{4.1}-2}{4.1}$

☐ C $\frac{\sqrt{4.1}-2}{0.1}$

☐ D $\sqrt{0.1}$

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Q10 · Rank 4 / Challenge · OPTIONAL

Find the coordinates of the point(s) at which the curve $y = x^5$ has gradient 405.

☐ A (3, 243) only

☐ B (3, 243) and $(-3, -243)$

☐ C $(81, 81^5)$ only

☐ D (3, 405) and $(-3, 405)$

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